Geotemporal Hydrodynamics
The Unified Fluid Theory

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Abstract

Geotemporal Hydrodynamics (GTH) proposes that spacetime is a macroscopic quantum superfluid rather than an empty geometric manifold. This framework models the vacuum as a 5D viscoelastic substrate with order parameter $\Psi = \rho e^{i\theta}$. This new report highlights four key phenomenological pillars of the theory based on recent visualizations.

1 Fluid Wake and Flow

Gravity emerges as the long-wavelength acoustic response of the substrate. The effective flow velocity incorporates both a Newtonian gravitational term and a macroscopic wake term parameterized by the Reynolds number (Re) and fluid density (ρ_0).

$$v = \sqrt{\frac{GM}{r} + \beta r} \tag{1}$$

where $\beta \propto Re \cdot \rho_0$.

2 GeoKnots (Topology)

Particles are not point objects but stable topological defects (GeoKnots) in the 5D viscoelastic substrate. The fundamental particle is modeled as a Trefoil Knot (3_1). Rest mass is defined as the localized kinetic bending energy of the substrate:

$$m_e \propto \frac{\sqrt{\rho_0}}{m_\psi^2} \tag{2}$$

Adjusting the scalar mass (m_ψ) changes the knot's thickness, while ρ_0 alters its overall scale.

3 Viscoelastic Relaxation

The substrate exhibits finite relaxation times (τ) when deformed, leading to viscoelastic damping. The fluid's response to rapid perturbations is governed by Israel-Stewart hydrodynamics, ensuring causality and providing a mechanism for ringdown dispersion/damping consistent with gravitational wave observations.

4 Rotation Curves

At galactic scales, the inverse energy cascade of the superfluid vacuum leads to anomalous rotation curves. While Newtonian predictions fall off as $v \propto 1/\sqrt{r}$, the GTH framework provides a non-Keplerian flat profile at large radii due to the βr term, naturally accounting for dark matter phenomenology.